

Introduction

In this assignment, you will apply the Markowitz mean-variance (MV) framework, tracking error variance (TEV) portfolio optimization framework, and the Arbitrage Pricing Theory (APT) to the analysis of portfolios that can be constructed with several ASX-listed stocks. You will also briefly investigate an extension of the MV framework to alternative optimization problems with different objective functions and/or constraints. Following a data pre-processing stage (“Part 0”), there are three main parts comprising the assignment:

Part 1 Construct the mean-variance (MV) optimal portfolios with and without a risk-free asset and the tracking error variance (TEV) minimizing portfolio with respect to the ASX 200 index.

Part 2 Analyze a k -factor model of the stock returns consisting of the first k principal components of the empirical covariance matrix of the centered returns time series and determine if the k -factor model is an APT model.

Part 3 Calculate optimal portfolios using other objective functions and/or constraints.

You will also be asked to answer several synthesis questions at the end of the assignment.

This assignment is worth 45% of your final mark for the subject. Marks are allocated to each part as follows:

Part	Marks
Part 1	14
Part 2	8
Part 3	18
Synthesis	5

Instructions

1. All computational work and responses to the questions in this assignment must be performed on a **Jupyter notebook**.
2. Your submission consists of two files:
 - (a) One **Jupyter notebook file** (in .ipynb format) for the assignment;
 - (b) One **.html file**, which is the Jupyter notebook exported as an HTML file showing the required outputs.
3. Clearly label all workings with the part and number of the question addressed by your work or code. Annotate your codes so the reader knows what is being done and which question/sub-question is being addressed in any cell/line (or cluster) of code.
4. Whenever applicable, use Markdown cells for your answers to the questions. Whenever a numerical answer must be reported or displayed, make sure that your notebook file shows the required output.
5. You are allowed to use generative AI to assist you in the completion of this task. You must disclose your use of any assistive tools in an Acknowledgements section at the end of your report. If you did not use generative AI in the preparation of your submission, you must say so in the Acknowledgment section.
6. **Failure to follow any of the aforementioned instructions will result to a penalty of 5 marks.**

Assumptions

These are in addition to the assumptions required for the Markowitz analysis, tracking error variance analysis, and the APT.

1. Assume there exists a risk-free asset whose return is $r_0 = 10\%$ per annum (i.e. annualized).
2. For all portfolio optimization analyses where a target return is required, the investor seeks a target expected return of 50% per annum (annualized) on their portfolio.

Part 0: Data Preparation

1. Load the data set `ASX200ClosePrices.csv`. This file contains daily close prices of the ASX 200 index (in the last column) and 191 of the constituent stocks of the ASX 200 covering the period from 1 January 2022 to 22 December 2023.
2. Compute the daily percentage returns of each stock and the index.
3. Determine the annualization factor

$$\frac{\text{no. of observed close prices}}{\text{no. of calendar days between start and end dates}/365}.$$

4. Calculate the mean and covariance matrix of the returns of the stocks and the mean and variance of the returns of the index. Using the annualization factor, express the means and covariances in an annual basis.
5. Sort the stocks by the Sharpe ratio $SR_i := \frac{E[R_i] - r_0}{\sqrt{\text{Var}(R_i)}}$ in descending order. Report the 15 stocks with the highest Sharpe ratio.
6. Calculate the empirical mean and covariance matrix of the (annualized) returns of the top 15 stocks identified in the previous step. These shall be one of the main inputs in the subsequent analyses.

Hint: If you followed the process above correctly, then the Sharpe ratio you should get for AGL Energy Limited (ticker: `AGL.AX`) is 0.493500.

For the rest of the assignment, we will assume that the top 15 stocks (in terms of the Sharpe ratio) comprise the investment universe or the set of risky assets we can invest in. Denote this set of risky assets/stocks by $\mathcal{S}$.

Part 1: Markowitz Mean-Variance and Tracking Error Variance Optimization

Markowitz Mean-Variance (MV) Analysis

1. Determine the mean-variance optimal (MV-optimal) portfolio with no risk-free asset ϕ_{NRF} for the given target expected return. Report the mean return and the standard deviation of the return of ϕ_{NRF} .
[1 mark]
2. Determine the MV-optimal portfolio with a risk-free asset allocation φ_{WRF} for the given target expected return. Report the mean return and the standard deviation of the return of φ_{WRF} .
[1 mark]
3. Briefly compare and discuss the MV-optimal portfolios with and without the risk-free asset in terms of composition and overall riskiness. Does either portfolio demand a large long or short position in any of the risky assets? Comment on whether there are any challenges in implementing these optimal portfolios in practice.
[2 marks]

- Construct the MV efficient frontier, its asymptote, and the capital market line associated to this investment universe for the range $m \in [-0.5, 1.0]$. Display these, along with the risk-return representation of ϕ_a , $\omega_{a,b}$, φ_τ , ϕ_{NRF} , φ_{WRF} , and the ASX 200 index, on a plot on the risk-return space.

Is the ASX 200 index efficient in this investment universe? Comment further on the risk-return characteristics of the ASX 200 using the plot you generated.

[2 marks]

Tracking Error Variance (TEV) Analysis

- For each stock in $\mathcal{S}$, determine its beta with respect to the ASX 200. Report the estimated betas graphically using a scatter plot whose x -axis is the stock's beta and whose y -axis is the stock's expected excess return. Would you expect the points to be perfectly aligned? Explain why or why not.

[2 marks]

- For the given target expected return, determine the composition of the TEV-optimal portfolio ϕ_{TEV} . Report the mean and standard deviation of its return. What is the TEV achieved by this portfolio?

Does the portfolio demand a large long or short position in any of the risky assets? Comment on whether there are any challenges in implementing this portfolio in practice.

[2 marks]

- Report the beta of the TEV-optimal portfolio with respect to the ASX 200 and compare this to the beta of the MV-optimal portfolios (with and without a risk-free asset). Briefly discuss what these betas imply regarding the riskiness of the optimal portfolios relative to the ASX 200.

[2 marks]

- Construct the TEV efficient frontier for this investment universe. Add the TEV efficient frontier and the risk-return representation of ϕ_{TEV} to the earlier visualization in the risk-return space and display the updated plot.

Discuss the risk-return representation of the ASX 200 relative to the TEV frontier. In this investment universe, does the ASX 200 lie on the TEV frontier? Explain why or why not.

[2 marks]

Part 2: k -Factor Models and Arbitrage Pricing Theory

- Construct the centered returns time series data for the risky assets in $\mathcal{S}$ and construct the covariance matrix $\hat{\Sigma}_c$ of the centered returns. **Do not annualize $\hat{\Sigma}_c$ for this analysis.** Determine the singular value decomposition (SVD) of $\hat{\Sigma}_c$.

What is the total variance or inertia of the sample? Display the cumulative proportion of the total variance explained by each principal component of $\hat{\Sigma}_c$.

[2 marks]

- Using `scikit-learn`, implement a principal component analysis (PCA) of the centered returns. Using the first $k = 5$ principal components, construct the k -factor model $R = A + BF + \epsilon$ implied by the PCA.

Display the coefficients A and B of the k -factor model and confirm that the variances of the factors coincide with the first k eigenvalues of $\hat{\Sigma}_c$ and that the error term has a zero mean vector.

[2 marks]

3. Is the PCA k -factor model an APT model? Justify and explain your answer by showing the necessary calculations.

[2 marks]

4. Compare the mean and covariance matrix of the risky asset returns obtained from the k -factor model and those obtained directly from the data (in Part 0). **Note that the covariance matrix and mean vector obtained using the k -factor model must be annualized.**

Do you expect your results in Part 1 to change significantly if you were to use the mean and covariance matrix obtained from the k -factor model? Explain your answer.

[2 marks]

Part 3: Alternative Portfolio Optimization Criteria

In this part, you will solve alternative portfolio optimization problems which may not necessarily have a closed-form solution. Examples of alternative portfolio optimization problems include the mean-variance analysis with additional constraints (as was illustrated in class) and portfolio optimization with different objective functions. You are tasked with solving these portfolio optimization problem using `scipy.optimize`, which is a general-purpose optimizer.

Example

Shown below is the (numerical) solution, using `scipy.optimize`, of the mean-variance portfolio optimization problem (with risky assets only) with a long-only constraint and a maximum weight constraint

$$\min_{\phi \in \mathbb{R}^d} \phi^\top \Sigma \phi \quad \text{subject to} \quad \phi^\top M = \bar{m}, \quad \phi^\top \mathbf{1}_d = 1, \quad 0 \leq \phi \leq 20\%,$$

where M is the mean return vector, Σ is the covariance matrix of returns, and the box constraint $0 \leq \phi \leq 20\%$ is understood componentwise.

```
1 import numpy as np
2 from scipy.optimize import minimize
3
4 # Inputs (given)
5 # M - mean return vector
6 # Sigma - covariance matrix of returns
7 target_ret = 0.30
8
9 # Objective function - to be MINIMIZED
10 def port_var(phi):
11     return phi.T @ Sigma @ phi
12
13 # Equality constraint - target mean return
14 def cons_target_ret(phi):
15     return (phi.T @ M - target_ret)
16
17 # Equality constraint - investment portfolio
18 def cons_inv_port(phi):
19     return (np.sum(phi) - 1)
20
21 # Inequality constraint - maximum weight
22
23 # Constraint dictionary
24 cons = ({'type' : 'eq', 'fun' : cons_inv_port},
25         {'type' : 'eq', 'fun' : cons_target_ret})
26
27 # Bounds on decision variables (componentwise constraints)
28 bnds = [(0, 0.20) for _ in range(len(M))]
29
30 # Optimization
31 phi_init = np.ones(len(M)) / len(M)
32 mv_cons_out = minimize(port_var, phi_init,
```

```

33         bounds = bnds, constraints = cons)
34
35 # Constrained mean-variance optimal portfolio composition
36 mv_cons_out.x

```

Listing 1: Python code to solve the constrained mean-variance portfolio optimization

The codes above can be modified to accommodate other objective functions and/or constraints for the portfolio optimization.

Questions

Using the basic structure of Listing 1, solve the following alternative portfolio optimization problems.

1. Consider the constrained mean-variance portfolio optimization problem

$$\min_{\phi \in \mathbb{R}^d} \phi^\top \Sigma \phi \quad \text{subject to} \quad \phi^\top M = \bar{m}, \quad \phi^\top \mathbf{1}_d = 1, \quad \phi^\top \beta_{\text{ASX}} = 0, \quad \max_i |\phi_i| \leq 0.50,$$

where β_{ASX} is the vector of betas of the stocks in $\mathcal{S}$ with respect to the ASX 200 index (computed in Part 1 TEV Task 1). Determine the composition, mean return, and standard deviation of the return of the portfolio ϕ_{MN} which solves the problem above. What are the practical implications of the final two constraints above?

[3 marks]

2. Find the portfolio ϕ_{MSR} that maximizes the Sharpe ratio

$$\max_{\phi \in \mathbb{R}^d} \frac{\phi^\top M - r_0}{\sqrt{\phi^\top \Sigma \phi}} \quad \text{subject to} \quad \phi^\top \mathbf{1}_d = 1, \quad \phi \geq 0.$$

Display the composition, mean return, and standard deviation of the return of ϕ_{MSR} .

[3 marks]

3. The Kelly portfolio is the portfolio ϕ_K which maximizes the (expected) portfolio growth rate, which can be shown in this context is given by $\mathbb{E}[\ln(1 + \phi^\top R)]$, where R is the random vector of returns of the risky assets over the investment period (Luenberger, 2014, Chapter 18). A mean-variance approximation of the Kelly criterion is given by

$$\mathbb{E}[\ln(1 + \phi^\top R)] \approx \phi^\top M - \frac{1}{2}(\phi^\top M)^2 - \frac{1}{2} \frac{\phi^\top \Sigma \phi}{1 + 2\phi^\top M}.$$

Find the portfolio ϕ_K that maximizes the Kelly criterion (using its mean-variance approximation), i.e.

$$\begin{aligned} \max_{\phi \in \mathbb{R}^d} \quad & \phi^\top M - \frac{1}{2}(\phi^\top M)^2 - \frac{1}{2} \frac{\phi^\top \Sigma \phi}{1 + 2\phi^\top M} \\ \text{subject to} \quad & \phi^\top \mathbf{1}_d = 1, \quad \phi \geq 0. \end{aligned}$$

Display the composition, mean return, and standard deviation of the return of ϕ_K .

[3 marks]

4. Optimal portfolios can also be determined based on the expected utility of the random portfolio return, $\mathbb{E}[U(\phi^\top R)]$, where U is a utility function. From this perspective, the Kelly criterion is the special case where the utility function is the log utility $U(x) = \ln(1 + x)$. Given a concave utility function U , the expected utility has a mean-variance approximation

$$\mathbb{E}[U(\phi^\top R)] \approx U(\phi^\top M) + \frac{1}{2} U''(\phi^\top M) \phi^\top \Sigma \phi.$$

Assuming the investor has an exponential utility function $U(x) = 1 - e^{-\delta x}$ (use $\delta = 10$), find the portfolio ϕ_{EU} that solves the problem

$$\begin{aligned} \max_{\phi \in \mathbb{R}^d} \quad & U(\phi^\top M) + \frac{1}{2} U''(\phi^\top M) \phi^\top \Sigma \phi \\ \text{subject to} \quad & \phi^\top \mathbf{1}_d = 1, \quad \phi \geq 0. \end{aligned}$$

Display the composition, mean return, and standard deviation of the return of ϕ_{EU} .

[3 marks]

5. Add the risk-return representation of ϕ_{MN} , ϕ_{MSR} , ϕ_K , and ϕ_{EU} to the efficient frontier plots you generated in Part 1 TEV Task 4.

[2 marks]

6. Calculate and display the Sharpe ratio and diversification ratio for ϕ_{NRF} , ϕ_{WRF} , ϕ_{TEV} , and the four optimal portfolios calculated in this section. Compare and contrast these portfolios based on these metrics.

[4 marks]

Part 4: Synthesis

Provide a brief critical assessment (two to four paragraphs in length) of the portfolio optimization criteria used in Parts 1 and 3. You must address the following points in your discussion:

[5 marks]

- Is it sufficient to base portfolio allocations only the mean and covariances of risky asset returns and, in the case of the TEV analysis, the beta of the risky assets with respect to a benchmark portfolio or index?
- What are the advantages and disadvantages of using the alternative portfolio optimization criteria in Part 3? Comment on the practicality of the alternative objective functions and/or constraints imposed.
- What other characteristics of the financial market or investment universe should be considered when optimally constructing portfolios? Explain why these characteristics are significant.

References

Luenberger, D. G. (2014). *Investment Science*. Oxford University Press, New York, USA, 2nd edition.